

Multivariate Normal Distribution

W02-L02

1. Density of Normal Distribution

$$X \sim N_p(\mu, \Sigma); \quad \Sigma > 0$$

$$f_X(x) = (2\pi)^{-p/2} |\Sigma|^{-1/2} \exp\left\{-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right\}, x \in \mathbb{R}^p$$

2. What happens if $|\Sigma| = 0$, $\Sigma \geq 0$?

Example

$$\begin{pmatrix} x \\ y \end{pmatrix} \sim N_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\right)$$

$$P(X = Y) = 1$$

$$a = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad (1 \quad -1) \begin{pmatrix} x \\ y \end{pmatrix} \sim N(0, 0).$$

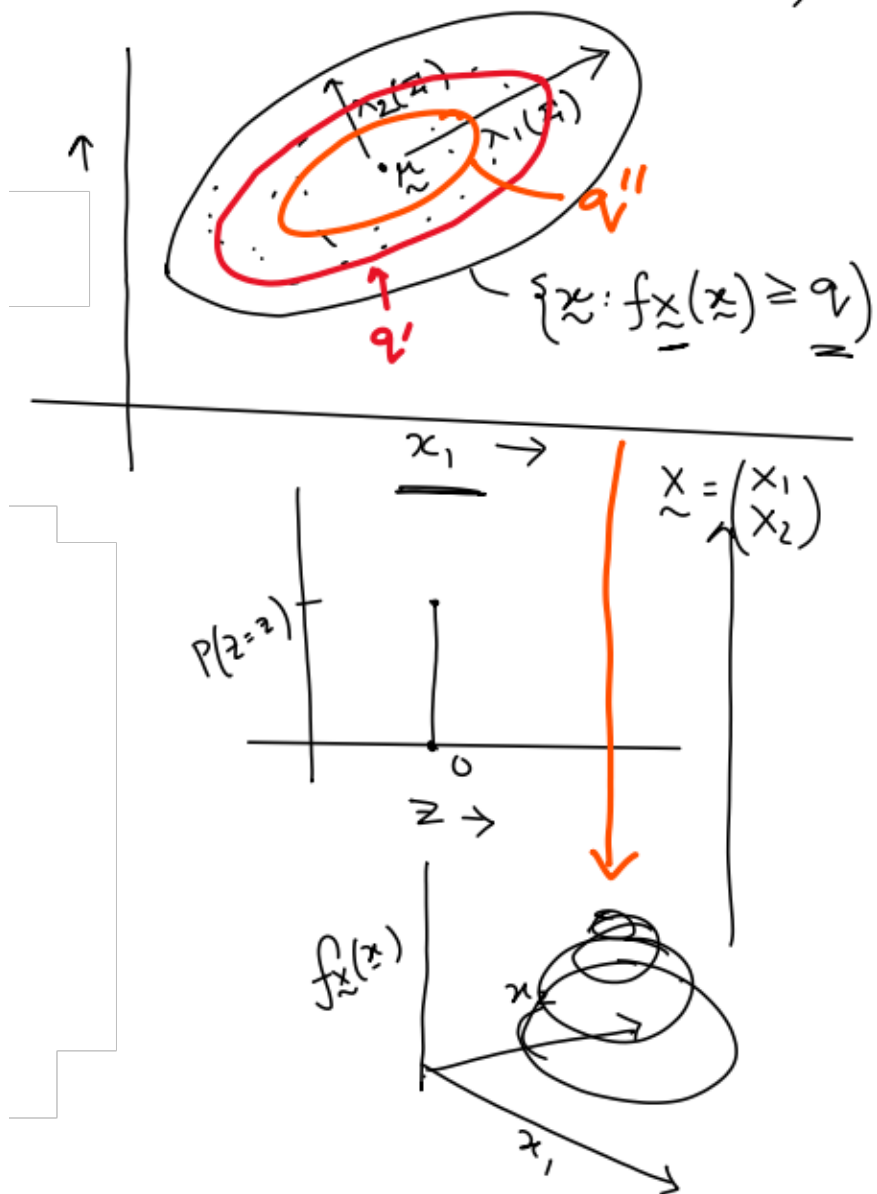
Let $Z = (1, -1) \begin{pmatrix} x \\ y \end{pmatrix}$. Then

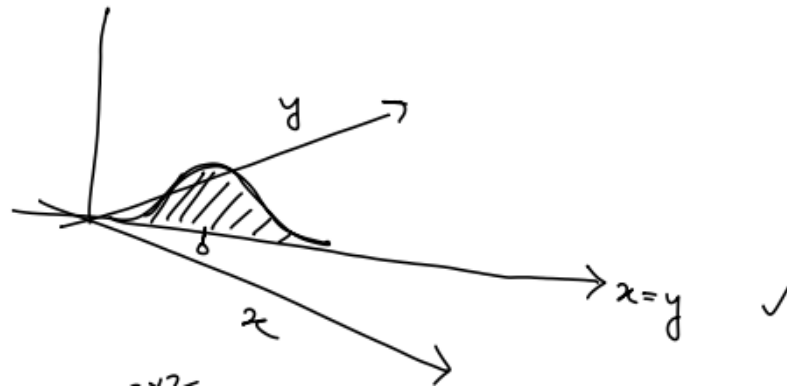
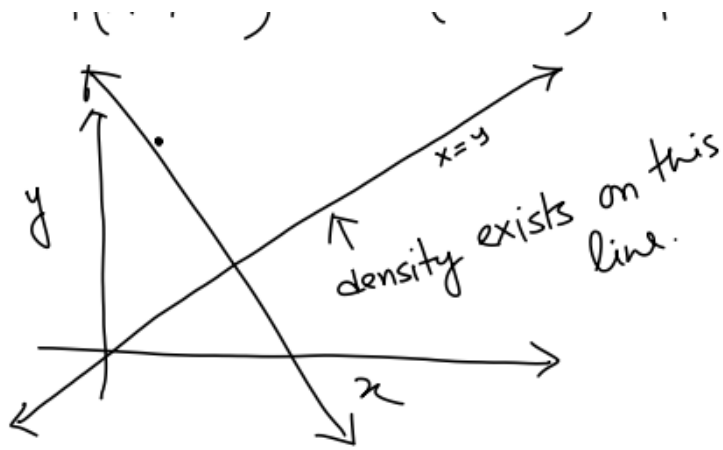
$$P(X - Y = 0) = P(Z = 0) = 1.$$

In general, if Σ is singular, then the density will exist on the line $\Sigma y = 0$ ($y \neq 0$).

$$y^T X \sim N(y^T \mu, y^T \Sigma y)$$

$$P(y^T X = 0) = 1$$





3. Independence of Two Linear Forms

$$X \sim N(\mu, \Sigma); \quad \Sigma > 0, \quad \text{when } AX \perp\!\!\!\perp BX.$$

$$\begin{bmatrix} Y \\ Z \end{bmatrix} = \begin{bmatrix} A \\ B \end{bmatrix} X = CX \sim N(C\mu, C\Sigma C')$$

When is $\text{Cov}(Y, Z) = 0$?

Answer: $A\Sigma B' = 0$.

Hence

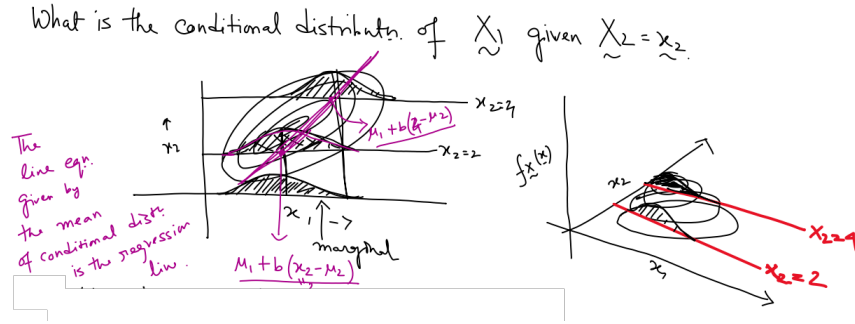
$$A\Sigma B' = 0 \iff AX \perp\!\!\!\perp BX.$$

4. Marginal and Conditional Distributions

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim N\left(\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}\right)$$

$$X_2 = [0 \ I_{p_2}]X \sim N(\mu_2, \Sigma_{22}).$$

Question. What is the conditional distribution of X_1 given $X_2 = x_2$?



The line equation given by the mean of the conditional distribution is the **regression line**:

$$\mu_1 + b(x_2 - \mu_2).$$

To obtain this, we follow the steps below:

- We will find B such that $Z = X_1 - BX_2 \perp\!\!\!\perp X_2$.

As $Z \perp\!\!\!\perp X_2$, the conditional distribution of Z given $X_2 = x_2$ is the same as the marginal distribution.

- Using this fact and the marginal distribution of Z , we will find the conditional distribution of $Z + BX_2 (=X_1)$ given $X_2 = x_2$.

Step (i)

$$Z = [I_{p_1} \ -B] \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, \quad X_2 = [0 \ I_{p_2}] \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

$$[I_{p_1} \ -B] \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{bmatrix} 0 \\ I_{p_2} \end{bmatrix} = 0$$

$$\iff [\Sigma_{11} - B\Sigma_{21} \quad \Sigma_{12} - B\Sigma_{22}] \begin{bmatrix} 0 \\ I_{p_2} \end{bmatrix} = 0$$

$$\iff \Sigma_{12} - B\Sigma_{22} = 0$$

$$\iff B = \Sigma_{12}\Sigma_{22}^{-1}$$

Therefore

$$Z = X_1 - \Sigma_{12}\Sigma_{22}^{-1}X_2$$

$$Z \perp\!\!\!\perp X_2$$

$$Z \sim N(\mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2, \Sigma_{11.2})$$

Step (ii)

Given $X_2 = x_2$,

$$\begin{aligned} & [\Sigma_{11} - B\Sigma_{21} \quad \Sigma_{12} - B\Sigma_{22}] \begin{bmatrix} I_{p_1} \\ -B' \end{bmatrix} \\ &= \Sigma_{11} - B\Sigma_{21} - \Sigma_{12}B' + B\Sigma_{22}B' \\ &= \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} + \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22}\Sigma_{22}^{-1}\Sigma_{21} \\ &= \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}. \end{aligned}$$

where

$$\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} \quad (\because \Sigma_{12} = \Sigma_{21}^T).$$

Hence

$$Z + \Sigma_{12}\Sigma_{22}^{-1}X_2 \sim N(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2), \Sigma_{11.2})$$

5. Distribution of Quadratic Forms

Idempotent matrix

A is idempotent if $A^2 = A$.

A is symmetric idempotent $\iff$ the eigenvalues of A are 1 with multiplicity $\rho(A)$ [rank of A] and 0 with multiplicity $p - \rho(A)$.

Property: $\rho(A) = \text{tr}(A)$.

Result

Let $Y \sim N(0, I_p)$ and let A be a symmetric matrix.

If A is idempotent then

$$Y^T A Y \sim \chi_{(q)}^2 \quad \text{where } q = \rho(A).$$

Proof

$$A = U \begin{pmatrix} I_q & 0 \\ 0 & 0 \end{pmatrix} U^T = \begin{bmatrix} U_1 & U_2 \end{bmatrix} \begin{pmatrix} I_q & 0 \\ 0 & 0 \end{pmatrix} \begin{bmatrix} U_1^T \\ U_2^T \end{bmatrix} = U_1 U_1^T$$

$$U U^T = U^T U = I$$

$$\begin{bmatrix} U_1^T \\ U_2^T \end{bmatrix} \begin{bmatrix} U_1 & U_2 \end{bmatrix} = \begin{bmatrix} U_1^T U_1 & U_1^T U_2 \\ U_2^T U_1 & U_2^T U_2 \end{bmatrix} = \begin{bmatrix} I_q & 0 \\ 0 & I_{p-q} \end{bmatrix}$$

$$\Rightarrow \begin{cases} U_1^T U_1 = I, \\ U_1^T U_2 = 0, \quad U_2^T U_2 = I. \end{cases}$$

Now

$$\mathbf{Y}^T A \mathbf{Y} = (U_1^T \mathbf{Y})^T (U_1^T \mathbf{Y}) \quad \text{where } U_1^T \mathbf{Y} \sim N_q(0, I)$$

$$= \|\mathbf{Z}\|^2, \quad \mathbf{Z} \sim N_q(0, I)$$

$$= \sum_{i=1}^q Z_i^2 \sim \chi_{(q)}^2 \quad \Longleftrightarrow \quad Z_i \stackrel{\text{iid}}{\sim} N(0, 1).$$

Example

$$\mathbf{Y} \sim N(0, \sigma^2 I), \quad S_y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

$$\frac{(n-1)S_y^2}{\sigma^2} \sim \chi_{(n-1)}^2$$

$$(n-1)S_y^2 = \sum_{i=1}^n Y_i^2 - n\bar{Y}^2, \quad n\bar{Y} = \sum_{i=1}^n Y_i = \mathbf{Y}^T \mathbf{1}$$

$$\begin{aligned} (n-1)S_y^2 &= \mathbf{Y}^T \mathbf{Y} - \frac{n}{n^2} \mathbf{Y}^T \mathbf{1} \mathbf{1}^T \mathbf{Y} \\ &= \mathbf{Y}^T \left(I - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) \mathbf{Y}. \quad \left(I - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) = A \end{aligned}$$

$$A^2 = A \quad (\text{verify})$$

$$\text{tr}(A) = \rho(A) = q = n-1$$

$$\mathbf{Z} = \frac{1}{\sigma} \tilde{\mathbf{Y}} \sim N(0, I), \quad \mathbf{Z}^T A \mathbf{Z} = \frac{\tilde{\mathbf{Y}}^T A \tilde{\mathbf{Y}}}{\sigma^2} \sim \chi_{(n-1)}^2.$$